

Two Bounces and One Underdetermination: A Machine-Checked Formal Metaphysics of an Involutive Arena

Abstract

We develop a formal metaphysics for an *involutive arena*: a pointed carrier equipped with a fixed-point-free involution. Within this ambient, the two-point carrier with Boolean negation is initial and unique up to label swap; a Lawvere-style diagonal argument yields an incompleteness (“seal”) with no appeal to arithmetisation; and a single $\mathbb{Z}/2\mathbb{Z}$ -underdetermination appears in three faces—pole, address, and channel. Transferring losslessness from the arena to forward-composing microsteps is discharged by factoring finite injective endomaps as products of two involutions. Canonicity of the factorisation fails: the absolute position of the two “mirrors” is conventional. What is real is their exchange, which covers step reversal. We introduce a demarcation between *content* gauges and *packaging* gauges that keeps this non-uniqueness from inflating the dictionary’s single $\mathbb{Z}/2\mathbb{Z}$. The development is machine-checked in Lean 4 without Mathlib and without **sorry**. Physics appears only as optional dictionary certificates, not as a coronation of the metaphysics.

Keywords: formal metaphysics; involution; Lawvere diagonal; underdetermination; gauge; machine-checked mathematics.

1 Introduction

Mathematical philosophy often proceeds by isolating a small formal structure and reading it metaphysically. The present paper does that for one structure—an involutive self-map on a pointed carrier—and then asks how much of a larger “arena” can be derived, transferred, and honestly refused.

Three results organise the argument.

- (i) **Canon.** Among pointed involutive acts, $(\mathbf{Bool}, \neg)$ is initial. Fundamentality coincides with reading as negation; uniqueness holds up to label swap.
- (ii) **Seal.** No self-reading is point-surjective onto the predicates of its own level. Incompleteness is semantic (Lawvere), not arithmetised (Gödel).
- (iii) **One underdetermination, three faces.** Pole swap, address reversal, and channel swap are presentations of a single $\mathbb{Z}/2\mathbb{Z}$. Channel *labels* are conventional; channel *swap* is proved content: it covers step reversal.

A fourth result is the hinge between arena and dynamics. Spine acts are involutions; ledger microsteps compose forward. The interface demand is that global injectivity of a joint step U be the transfer of in-arena losslessness. That transfer is discharged by a classical finite fact: injective endomaps of $\mathbf{Fin} n$ factor as $\sigma \circ \phi$ with both factors involutions (“two-bounce”). Canonicity of (ϕ, σ) fails; the failure is not a second dictionary $\mathbb{Z}/2\mathbb{Z}$ but a *packaging* gauge—spent inside the discharge, never exported to consumers of U .

The metaphysical signature that emerges is relational throughout: absolute positions (which pole, which mirror, which channel label) are unreal; exchanges and relations are real. The paper’s contribution is to make that signature theorem-shaped, machine-checked, and demarcated against gauge inflation.

Scope and method. We present mathematics with a metaphysical reading. Continuum physics, host-logic incompleteness, and experiential claims are refused. Accompanying Lean declarations are cited by name; the paper is self-contained at the level of statements. The formal corpus uses Lean 4 with no Mathlib dependency and no `sorry`.

Positioning. Relative to Quinean underdetermination, the present $\mathbb{Z}/2\mathbb{Z}$ is narrower and harder: not “theory vs world,” but a proved two-element symmetry internal to the arena, with a proved exchange and a failed canonicity attempt for absolute labels. Relative to structural realism, the moral is similar—relations over absolutes—but obtained as a theorem package rather than a stance. Relative to formal metaphysics in the modal or mereological style, the primitives here are dynamical (an act) rather than static (worlds, parts).

Roadmap. Section 2 fixes the arena and the canon. Section 3 states the seal. Section 4 packages the one $\mathbb{Z}/2\mathbb{Z}$. Section 5 discharges the losslessness transfer and proves canonicity failure together with swap-as-reversal. Section 6 states the content/packaging demarcation. Section 7 records residue and refusals. Section 8 concludes.

2 The involutive arena and the canon

Definition 2.1 (Arena data). *Fix a type α and a map $a : \alpha \rightarrow \alpha$. Write*

$$\text{Step}(a)(x, y) :\Leftrightarrow y = a(x).$$

Call a a symmetric step when $\text{Step}(a)$ is a symmetric relation; equivalently, $a \circ a = \text{id}$ (an involution). Call a live when it has no fixed point: $\forall x. a(x) \neq x$. Call a lossless when it is injective.

Pointwise, an act either rests ($a(x) = x$), closes a two-cycle ($a(x) \neq x$ and $a(a(x)) = x$), or carries an asymmetric step ($a(a(x)) \neq x$). Erasure—distinct states with a common successor—is the pairwise defect of losslessness.

Why involutions? Because the ambient is chosen to exclude smuggled time and smuggled loss at once. An asymmetric step is already an arrow; an erasing step is already a collapse. Symmetric step is the equation that bans both. The metaphysics is therefore conditional: *if* one works inside an involutive arena, *then* the theorems below apply. The facticity of that “if” remains residue (Section 7).

Definition 2.2 (Pointed unoriented acts). *Objects are triples (α, a, z) with a a symmetric step and $z : \alpha$ a seed. Morphisms are pointed equivariant maps.*

Theorem 2.3 (Canon / initiality). *In the ambient of pointed unoriented acts, $(\text{Bool}, \neg, \text{true})$ is initial. An act is fundamental precisely when it is live, symmetric, and generated from a seed by the two-cycle reading; every fundamental reads as $\neg$; any two fundamentals correspond uniquely up to the pointed reading.*

Thus fundamentality is not an extra primitive: it is initiality plus the reading theorems. The excluded fixed point (“void”) is just liveness. Involutivity simultaneously banishes asymmetric arrow and erasure at the base: the same equation $a \circ a = \text{id}$ yields both.

Proposition 2.4 (In-arena losslessness). *If a is a symmetric step, then a is not erasing.*

This is the spine’s own losslessness theorem. It is *not* a second creed beside involutivity; it is a corollary. The metaphysical reading is accordingly modest: inside an involutive arena, collapse of distinct states under the act is impossible. What happens when one steps “upstairs” to forward-composing dynamics is the subject of Section 5.

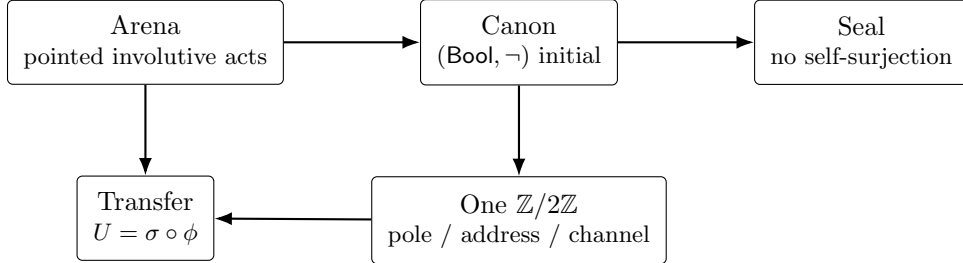


Figure 1: Dependency sketch. Canon and seal live on the arena; the single $\mathbb{Z}/2\mathbb{Z}$ names its underdetermination; two-bounce transfers losslessness to forward steps without exporting factor choice.

3 Seal: incompleteness without arithmetisation

Lawvere’s fixed-point argument yields incompleteness from a Cartesian closed structure and a weakly point-surjective morphism, without coding syntax as numbers. The arena supplies a concrete instance.

Theorem 3.1 (Seal on a fundamental). *Let a be live. No map $f : A \rightarrow (A \rightarrow \text{Bool})$ is point-surjective: the diagonal dodge $x \mapsto \neg(f\ x\ x)$ is missed. Specialising to fundamental acts, every self-reading admits an escape.*

A companion result at corpus grain states that no self-reading of the formal corpus is universal for the predicates of its own level (**corpus_not_universal**): names are outnumbered by predicates; any completed self-portrait lags. Completeness-in-the-limit is a *policy* (a fair schedule of self-readings), not a theorem that the corpus catches itself.

Remark 3.2. *The seal is not a claim that Lean cannot prove its own consistency, nor a Gödel arithmetisation of the corpus. Those are refused as category errors for this project. The theorem is semantic and local to the arena’s self-readings.*

Metaphysically: an involutive arena that can name its own escapes cannot close those names into a total self-surjection. Forced growth (“emanation” as naming the dodge) is the positive face of the same fact. Concretely, one builds an ω -ladder: level 0 is the fundamental carrier; each successor adjoins a namer for the previous level’s dodge; history embeds injectively and conservatively; no level exhausts its successor. Duration, in this sense, is the order-type of self-escape. Direction along the ladder remains free ($\Box(\text{ascent}) \wedge \neg\Box(\text{direction})$): ascent is forced, which escape is named is not. The same pattern—forced relation, free absolute—returns in Sections 4 and 5.

4 One underdetermination, three faces

Theorem 4.1 (One $\mathbb{Z}/2\mathbb{Z}$, three faces). *There is a single two-element underdetermination with three named involutive projections:*

$$\text{pole swap (reading)} \cong \text{address \{fwd, rev\}} \cong \text{channel \{\varphi, \sigma\}}.$$

Each projection is an involution on a common $\mathbb{Z}/2\mathbb{Z}$ carrier.

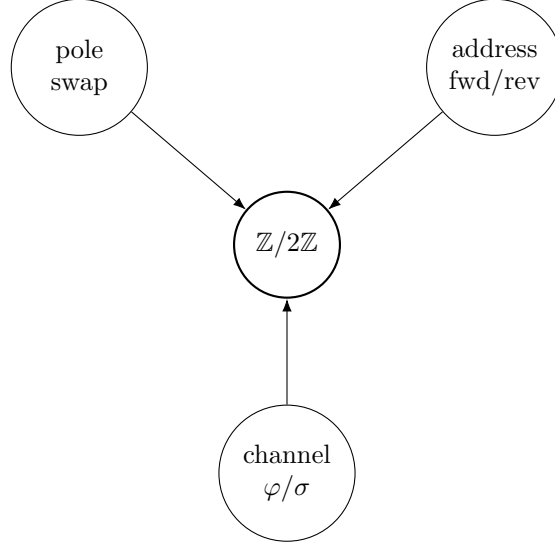


Figure 2: T-7: three named faces of one underdetermination. The faces are presentations; the group is stated once.

The philosophical load is carried by *stating the group once*. Repeating $\mathbb{Z}/2\mathbb{Z}$ in three vocabularies without identifying them would be concordance; identifying them without a theorem would be rhetoric. Theorem 4.1 is the identification.

Section 5 sharpens the channel face: absolute channel position is unreal (canonicity fails), while channel *exchange* is real (swap covers reversal). That is the same relational pattern the spine wears at $\neg\Box(\text{direction})$ and that cut-blind modular dynamics wear when period is global: absolutes are refused; relations survive.

5 Transferring losslessness: two bounces

Spine acts are involutions. A ledger microstep is a forward map $U : S \times E \rightarrow S \times E$. The interface demand (I-1) is that global injectivity of U be the import of Proposition 2.4 onto that forward dynamics.

Definition 5.1 (Two-bounce factorisation). *A two-bounce factorisation of $U : \alpha \rightarrow \alpha$ is a pair of involutions (ϕ, σ) with $U = \sigma \circ \phi$ pointwise.*

Theorem 5.2 (Converse). *If U admits a two-bounce factorisation, then U is lossless (injective). On a product carrier this is ledger injectivity.*

Theorem 5.3 (Existence on finite carriers). *Every injective $U : \text{Fin } n \rightarrow \text{Fin } n$ admits a two-bounce factorisation. The construction is cycle-wise reflection at a least-value basepoint, with $\sigma := U \circ \phi$.*

Existence plus converse discharges the interface shape: on finite carriers, injectivity and two-bounce form coincide. Toys (involutions as $U \circ \text{id}$; Boolean injections; a swap step) sit inside the same package.

5.1 Canonicity fails

Theorem 5.4 (Canonicity fails). *There exist an injective $U : \text{Bool} \rightarrow \text{Bool}$ and two two-bounce factorisations that disagree on ϕ (equivalently: $\neg$ factors as both $\neg \circ \text{id}$ and $\text{id} \circ \neg$). Moreover, on Fin3 two reflection-axis factorisations of a 3-cycle disagree and are not related by bounce-swap $(\phi, \sigma) \mapsto (\sigma, \phi)$.*

In the dihedral picture, a rotation determines the angle between its two mirrors, not their absolute position. Theorem 5.4 is that fact discretely: the step fixes the relation between the bounces; where the mirror pair sits is convention. Absolute channel position is unreal; relative channel structure is real.

5.2 Channel-swap is reversal

Non-uniqueness does not block a proved channel column. Section III never needed canonical factors; it needed a canonical *swap*.

Theorem 5.5 (Channel-swap is reversal). *Let $U = \sigma \circ \phi$ with ϕ, σ involutions. Then $\phi \circ \sigma$ is a two-sided inverse of U , and (σ, ϕ) is a two-bounce presentation of that reverse step. The construction is independent of which factorisation of U was chosen.*

Proof sketch. From $U = \sigma \circ \phi$ and the involution equations, $\phi(\sigma(U(x))) = \phi(\sigma(\sigma(\phi(x)))) = \phi(\phi(x)) = x$ and $U(\phi(\sigma(x))) = \sigma(\phi(\phi(\sigma(x)))) = \sigma(\sigma(x)) = x$. The swapped pair (σ, ϕ) satisfies $(\phi \circ \sigma) = \phi \circ \sigma$ definitionally as a two-bounce presentation of the reverse step. No uniqueness of (ϕ, σ) was used. $\square$

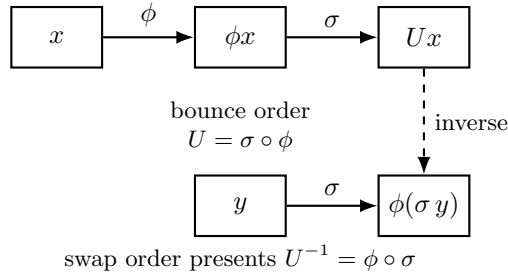


Figure 3: Two-bounce presentation and its swap. Labels (ϕ, σ) are conventional; exchanging them runs the step backwards.

On an involution, the reverse step equals U itself, so factor-swap re-presents the same step—the T-7-shaped reorder already visible in $\neg \circ \text{id}$ versus $\text{id} \circ \neg$. In general, channel-swap and address-swap (reading the step forward versus reverse) project to the same inversion $U \mapsto U^{-1}$. The corpus cannot say which mirror is which, and can prove that swapping them is running the step backwards.

6 Content gauges and packaging gauges

Gauge inflation is the natural hostile-reader objection once factorisation non-uniqueness is on the table: has a second $\mathbb{Z}/2\mathbb{Z}$ appeared beside T-7? The answer is no, once a demarcation is stated.

Definition 6.1 (Content vs packaging). *A gauge is content when some exported statement mentions the choice. A gauge is packaging when the choice is quotiented before anything downstream consumes it.*

Gauge	Locus	Class
Label / pole swap	canon readings	content
Address fwd/rev	recovery / arrow line	content
Channel swap	Theorem 5.5	content
Record gauge per tick	archive normal form	packaging
Factorisation gauge per step	Theorems 5.3–5.4	packaging

The ledger consumes U , never the pair (ϕ, σ) . Axis choice on a 3-cycle is spent entirely inside I-1 discharge. Record gauge appears only inside “up to record gauge” clauses. By contrast, pole, address, and channel swap survive into the dictionary line and the arrow identification, so they are content—and they are the *same* content, stated once as Theorem 4.1.

This demarcation has been used twice before being named (record gauge; factorisation gauge). Naming it makes the one- $\mathbb{Z}/2\mathbb{Z}$ claim hostile-reader-proof: the gauge grew, and it grew in a direction that never touches the dictionary.

7 Residue and refusals

Not everything nearby is a theorem, and the paper’s credibility depends on saying so.

Philosophical residue. The facticity of liveness—why there is a live act at all—is not discharged. Ambient adequacy (whether pointed unoriented acts are the right setting for fundamentality) remains contestable. These are residue, not hidden axioms of the theorems above.

Refused. Premise-free global laws of time; continuum spacetime and continuum Einstein equations as theorems of this corpus; host-logic incompleteness and Gödel arithmetisation of the formal text; experiential or qualia claims. Dictionary instantiations (physical models admitted by certificate) are existence proofs that the kernel symmetry has witnesses, not a derivation of the world from Bool.

Extension, not the thesis. A ledger layer (registration, capacity, discrete Einstein skeleton) and dictionary certificates sit above the arena. They illustrate transfer and constraint; they are not required for Theorems 2.3–5.5. The present paper isolates the metaphysical core.

8 Conclusion

An involutive arena yields a short list of hard theorems: a Boolean canon; a Lawvere seal; one $\mathbb{Z}/2\mathbb{Z}$ with three faces; a discharged losslessness transfer whose factorisation is non-canonical; and a proved identification of channel-swap with step reversal. The accompanying demarcation between content and packaging gauges prevents non-uniqueness from counting as a second underdetermination.

Taken together, the strengths are not merely additive. Initiality gives a unique fundamental *object*; the seal gives a negative *epistemic* bound on self-reading; T-7 gives a unique *underdetermination*; two-bounce gives a discharged *interface*; canonicity failure plus swap-as-reversal give a *relational* reading of the channel face that matches the spine’s refusal of absolute direction. The gauge demarcation is what makes the package stable under hostile reading.

The system keeps refusing absolutes at every level anyone proposes one—direction of ascent, position of mirrors, labels of channels—and the refusals were found by attempting

the opposite. That is the strongest kind of negative result available to formal metaphysics: not a slogan, but a failed canonicity proof, machine-checked, that promotes the exchange it could not eliminate.

Formal companion

Statements correspond to a Lean 4 development (no Mathlib, no `sorry`). Representative declarations: `canon_initial`, `symmetric_excludes_erase`, `seal_on_fundamental`, `corpus_not_universal`, `one_Z2_three_faces`, `factor_fin`, `canonicity_fails`, `channel_swap_is_reversal`. Axiom footprints are audited per declaration; the core arena results are axiom-free or depend at most on standard propositional extensionality / choice used in finite search.

References

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